Guide: Environment Setup for Huawei Cloud Development

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Contents

1 Provision an ECS Instance on Huawei Cloud

General Objective: Provision an Elastic Cloud Server (ECS) on Huawei Cloud that will host a complete coding stack environment: the oh-my-coding-maas-gateway (a LiteLLM-based proxy with load balancing, virtual keys, and Grafana observability), governance tooling, and multiple coding agents — with OpenCode as the primary one. By the end of this chapter you will have a running ECS instance with a public IP accessible over SSH.

Prerequisites:

- Active Huawei Cloud account with IAM credentials.
- Permission to create security groups, ECS instances, and EIPs.
- A key pair created in the Console, or permission to create one.
- A web browser (Chrome, Firefox, or Edge).

> **Info**
> This guide assumes the `sa-brazil-1` region. Adjust the region if your account is registered elsewhere.

1.1 Create the security group (sg-xgate)

Objective: Create a security group that allows SSH access on port 4444 from specific Huawei internal IPs only.

> **Info**
> We use port `4444` instead of the standard SSH port `22` because the Huawei corporate proxy blocks outbound connections on port 22. Port `4444` is allowed through the proxy and works reliably for SSH.

Step by step:

1. In the Console, go to **VPC** → **Security Groups** and click **Create Security Group**.
2. Name it `sg-xgate` and add the following inbound rules:

```
Rule 1:
  Protocol:    TCP
  Port:        4444
  Source:      119.8.89.193/32
  Description: SSH for Huawei internal user 1

Rule 2:
```

```
  Protocol:    TCP
  Port:        4444
  Source:      119.8.89.3/32
  Description: SSH for Huawei internal user 2

Rule 3:
  Protocol:    TCP
  Port:        4444
  Source:      119.8.193.3/32
  Description: SSH for Huawei internal user 3
```

3. Click **OK** to create the security group.

> **Tip**
> Each /32 rule restricts access to exactly one IP address. Only the three listed Huawei internal IPs can reach the ECS on port 4444 — no other inbound traffic is allowed.

1.2 Create the ECS instance

Objective: Launch a single ECS instance using the Console wizard, using the security group created above.

Step by step:

1. Log in to the Huawei Cloud Console and select the `sa-brazil-1` region in the top-right corner.
2. Open the service catalog and select **Elastic Cloud Server (ECS)**.
3. Click **Create ECS** and fill in the basic configuration:

```
Name:           dev-server-01
Flavor:         ac8.xlarge.2 (4 vCPU, 8 GB RAM) - recommended
Image:          Ubuntu 24.04 Server 64bit
Disk:           100 GB General Purpose SSD
VPC:            vpc-default
Subnet:         subnet-default
Security Group: sg-xgate (port 4444, Huawei internal IPs only)
```

4. Under **Billing Mode**, select **Pay-per-use**.

> **Tip**
> Pay-per-use bills by the hour with no long-term commitment. You only pay for the time the instance is running. This is ideal for development and testing — stop the instance when not in use to reduce costs.

5. Under **EIP** settings, configure the public IP:

```
EIP:            Auto-assign
Billing:        Pay by traffic
Bandwidth:      1000 Mbit/s
Release:        Yes - release EIP when ECS is deleted
```

6. Under **Login Configuration**, select **Key pair** as the authentication method. If you do not yet have a key pair, click **Create Key Pair** to generate one inside Huawei Cloud and download the `.pem` file.

   > **Info**
   > The key pair is the primary access method. Later, you can set a root password on the ECS (via the Console or after first login with the key) to have a second way to access the instance.

7. Click **Create Now** and wait for the status to change to **Running**.

   > **Important**
   > The EIP is auto-assigned and configured to be released when the ECS is deleted. If you need a fixed IP that persists across stop/start cycles, allocate a dedicated EIP and bind it manually after creation.

1.3 Verify the instance

On the ECS details page, copy the **EIP** field. This is the address you will use to connect in the next chapters.

2 Configure ECS for Remote Access

General Objective: Prepare the ECS instance for remote access by setting a root password via the Console and changing the SSH port from 22 to 4444. This is required because the Huawei corporate proxy blocks outbound connections on port 22.

Prerequisites:

- Chapter 1 completed — ECS instance running with a public IP.
- Access to the Huawei Cloud Console.

2.1 Set the root password

Objective: Use the Console to reset the root password of the ECS instance.

Step by step:

1. In the Console, navigate to **Elastic Cloud Server** and click on the instance `dev-server-01`.
2. In the ECS details page, click **More** → **Reset Password**.

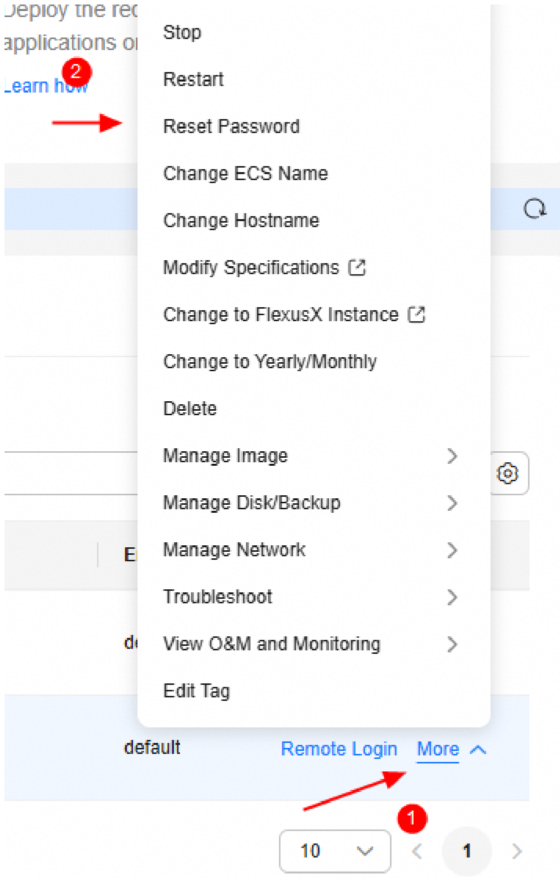

Figure 1: Reset Password option in the ECS details page.

3. Enter a strong root password and confirm. Click **OK**.
4. The ECS will restart automatically. Wait for the status to return to **Running**.

Important

Choose a strong password (12+ characters, mixed case, digits, symbols). This password enables remote login directly from the Console and serves as a second access method alongside the SSH key pair.

2.2 Change the SSH port to 4444

Objective: Log in via the Console remote login, edit `sshd_config`, and change the SSH port from 22 to 4444.

Step by step:

1. In the ECS details page, click **Remote Login** (top-right corner). A browser-based terminal opens.

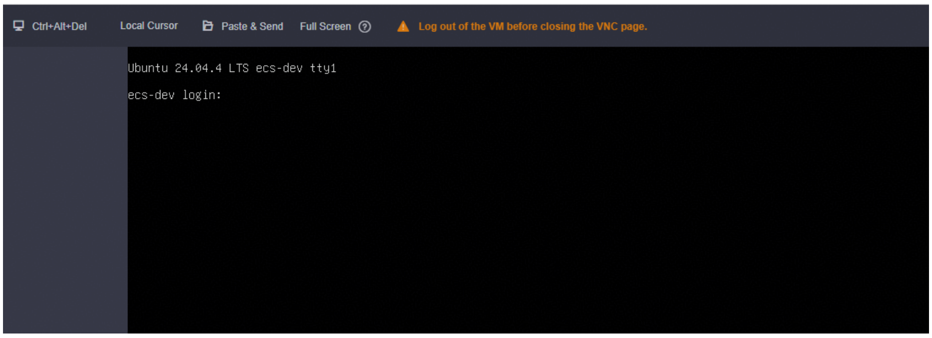

Figure 2: Remote Login button in the ECS details page.

2. Log in as `root` using the password set in the previous subsection.
3. Open the SSH daemon configuration file:

```
sudo vim /etc/ssh/sshd_config
```

4. Find the line `Port 22` (or `#Port 22`) and change it to:

```
Port 4444
```

5. Save and exit vim (`:wq`).
6. Restart the SSH daemon (or reboot the ECS):

```
sudo systemctl restart sshd
```

> **Tip**
> Alternatively, restart the ECS from the Console. The SSH daemon will start automatically on port 4444 after reboot.

> **Info**
> After this change, SSH connections must use `Port 4444` explicitly. The security group `sg-xgate` (created in Chapter 1) already allows inbound TCP 4444 from the authorized Huawei internal IPs.

2.3 Verify the configuration

Use the table below to confirm all steps were completed successfully before proceeding to the next chapter:

Step	What to check	Expected result
Security group	`sg-xgate` exists with TCP 4444 rules	3 inbound rules
ECS running	Instance `dev-server-01` status	**Running**
Root password	Remote Login works with password	Shell prompt
SSH port	`sshd_config` has `Port 4444`	Config saved
SSH restart	`sshd` restarted or ECS rebooted	Service active

Table 1: Pre-SSH verification checklist.

3 Connect via SSH with Proxy for VS Code

General Objective: Configure VS Code Remote-SSH to connect to the ECS instance through the Huawei corporate proxy (xgate) using `connect.exe` from Git for Windows.

Prerequisites:

- Chapter 2 completed — ECS configured with root password and SSH on port 4444.
- VS Code installed on your local Windows machine.
- The private key file (`.pem`) downloaded from Huawei Cloud.

> **Info**
>
> **Premises:**
> - The proxy is **xgate** (Huawei corporate proxy) at `proxy.huawei.com:8080`.
> - SSH runs on port `4444` (configured in Chapter 2).
> - Login as `root` using the key pair from Chapter 1.
> - `connect.exe` is used instead of `ncat.exe` — it ships with Git for Windows, so no extra installation is needed.

3.1 Install the Remote-SSH extension

Step by step:

1. Open VS Code on your local Windows machine.
2. Install the **Remote - SSH** extension (publisher: Microsoft):

   ```
   code --install-extension ms-vscode-remote.remote-ssh
   ```

3.2 Install Git for Windows

Objective: Install Git for Windows, which provides `connect.exe` — the proxy tunnel tool used by SSH `ProxyCommand`.

Step by step:

1. Download the Git for Windows installer from git-scm.com/download/win.
2. Run the installer with the default settings. Click **Next** through each step — the defaults are correct for our use case.

3. After installation, verify that `connect.exe` is available:

```
"C:\Program Files\Git\mingw64\bin\connect.exe" --help
```

4. If the command above shows help text, the installation is correct. If not, check whether you installed 32-bit Git (the path would be `mingw32` instead of `mingw64`).

> **Info**
> `connect.exe` is included with Git for Windows and supports HTTP proxy tunneling via the `-H` flag. It is a lightweight alternative to `ncat.exe` (from Nmap) and requires no separate installation.

3.3 Configure the SSH config file

Objective: Create an SSH config entry that tunnels through the Huawei proxy using `connect.exe` from Git.

Step by step:

1. Open or create the SSH config file on your Windows machine:

```
notepad %USERPROFILE%\.ssh\config
```

2. Add the following configuration (replace `<EIP>` with the ECS public IP and `<host-name>` with a name of your choice):

```
Host <host-name>
    HostName <EIP>
    User root
    Port 4444
    IdentityFile "C:\Users\user\.ssh\KeyPair-opencode.pem"
    ProxyCommand "C:\Program Files\Git\mingw64\bin\connect.exe" -H
    proxy.huawei.com:8080 %h %p
    ServerAliveInterval 60
    ServerAliveCountMax 3
```

> **Info**
> Adjust the `IdentityFile` path to match the location where you saved the `.pem` key file. The `connect.exe` path assumes a 64-bit Git installation — if you use 32-bit Git, change `mingw64` to `mingw32`.

> **Tip**
> The `ServerAliveInterval` and `ServerAliveCountMax` settings keep the connection alive through the proxy, which may drop idle connections.

3.4 Connect to the ECS via VS Code

Step by step:

1. Press `F1`, type **Remote-SSH: Connect to Host...**, and select your host name from the list.
2. VS Code opens a new window connected to the remote instance. Click **Open Folder** and select `/root` (or your working directory).
3. Verify the connection by opening a terminal in VS Code (`Ctrl+``) and running:

```
uname -a
df -h
```

4 Install the MaaS Coding Gateway

General Objective: Install the oh-my-coding-maas-gateway on the ECS instance. This is a self-hosted LiteLLM proxy that routes Huawei ModelArts MaaS models to coding tools (opencode, Codex CLI, Claude Code, Pi agent) with load balancing, virtual keys, and Prometheus + Grafana observability.

Prerequisites:

- Chapter 3 completed — VS Code connected to the ECS via Remote-SSH.
- Docker and Docker Compose available (the installer handles this).
- A Huawei ModelArts MaaS API key.

> **Info**
> The gateway repository is at wallacelw/oh-my-coding-maas-gateway. It supports 4 models: `glm-5.2`, `glm-5.1`, `deepseek-v4-pro`, and `deepseek-v4-flash`.

4.1 Run the bootstrap installer

Objective: Install the gateway using the one-command bootstrap script.

Step by step:

1. In the VS Code remote terminal, run the bootstrap script:

   ```
   curl -fsSL https://raw.githubusercontent.com/wallacelw/oh-my-coding-maas-gateway/main/scripts/bootstrap.sh | bash
   ```

2. The installer will:
   - Prompt for your Huawei MaaS API key.
   - Show an interactive menu to select which coding tools to install (opencode, Codex CLI, Claude Code, Pi agent).
   - Auto-install prerequisites (Docker, Node.js, etc.).
   - Deploy the proxy + observability stack (LiteLLM, PostgreSQL, Prometheus, Grafana).
   - Configure each tool with its own virtual key.
   - Run validation and offer to install the companion skill.

   > **Important**
   > Estimated time: ~5 min for a fresh install, ~2 min for an upgrade. If you are behind a proxy, ensure `http_proxy` and `https_proxy` environment variables are set before running the script.

4.2 Verify the installation

Step by step:

1. Check that the proxy is running:

   ```
   curl http://localhost:4000/health
   ```

2. Open the Grafana dashboard at `http://localhost:3000` to monitor usage and metrics.
3. Open the LiteLLM Admin UI at `http://localhost:4000/ui` to manage virtual keys and model routing.
4. Launch your coding tool:

   ```
   opencode          # or:  codex  or:  claude --bare  or:  pi
   ```

> **Tip**
> To upgrade the gateway later, re-run the same bootstrap command. It detects existing installs, compares versions, and pulls updates — all secrets and data are preserved.

5 Install the Huawei Document Templates Project

General Objective: Clone the Huawei document templates repository on the ECS instance, run the setup script, and verify that the LaTeX toolchain and the coding agent skill are ready for use. The script assumes a fresh Ubuntu install — it installs all dependencies from scratch and configures VS Code at the user level.

Prerequisites:

- Chapter 4 completed — MaaS gateway installed and running.
- Git installed (`sudo apt install git`).
- The repository URL for the Huawei document templates project.
- A fresh Ubuntu 24.04 instance (no prior LaTeX installation required).

Info

The `install.sh` script is idempotent — it is safe to run multiple times. It merges VS Code settings rather than overwriting them, and `apt-get install` skips packages that are already present.

5.1 Clone and set up the project

Step by step:

1. Clone the repository:

```
cd ~
git clone <repo-url> huawei-doc-templates
cd huawei-doc-templates
```

2. Run the setup script (installs XeLaTeX, latexmk, fonts, the coding agent skill, and configures VS Code LaTeX Workshop at the user level):

```
./install.sh
```

Important

The script runs `sudo apt-get install` for TeX Live packages. If you do not have sudo access, ask your administrator to install the packages listed in `install.sh`.

3. Verify the LaTeX toolchain:

```
xelatex --version
latexmk --version
```

5.2 Compile the sample guides

Step by step:

1. Compile the Portuguese sample:

```
cd examples/guide/pt
latexmk main.tex
```

2. Compile the English sample:

```
cd ../en
latexmk main.tex
```

3. Verify the PDFs were generated:

```
ls -l examples/guide/pt/main.pdf
ls -l examples/guide/en/main.pdf
```

5.3 Create your first guide with the skill

Step by step:

1. In the VS Code remote terminal, launch your coding agent:

```
cd ~/huawei-doc-templates
opencode          # or: codex, claude --bare, pi
```

2. Run the guide skill:

```
/skill huawei-template-guide
```

3. The skill will ask for:
   - **Title** — e.g. "Provisioning an OBS Bucket"
   - **Language** — Portuguese or English
   - **Project name** — used as the folder name
   - **Location** — where to create the project folder
4. The skill generates the `.tex` file, creates a `.latexmkrc` with the correct `TEXINPUTS`, compiles the document, and reports the page count.
   Ready

> **Info**
> The skill is auto-discovered via `opencode.json` which registers `templates/` as a skill discovery path. No manual skill installation is needed when working inside the repo.

5.4 Write content using template commands

The template provides commands for common guide elements:

Headings (automatic numbering, H1 starts on a new page):

```
\section{Chapter Title}          % H1: giant number + title + rule
\subsection{Section Title}        % H2
\subsubsection{Subsection Title}  % H3
```

Objectives block (closes with a horizontal rule):

```
\begin{objectives}
  \generalobjective{...}
  \prerequisites
  \begin{itemize}
    \item ...
  \end{itemize}
\end{objectives}
```

Callout boxes and inline commands:

```
\begin{warning} ... \end{warning}     % amber warning box
\begin{tip}  ... \end{tip}      % green tip box
\begin{infobox} ... \end{infobox} % blue info box
```

```
\inlinecode{inline code}                % inline monospace
\image{assets/screenshot.png}    % centered image
\note{Italic observation.}        % note paragraph
\weblink{https://...}{link text}  % blue clickable link
\badge{New}                       % inline red label
```

6 Clean Up

General Objective: Remove the resources created in this guide to avoid ongoing charges on your Huawei Cloud account.

Step by step:

1. In the Console, navigate to **Elastic Cloud Server**.
2. Select the instance `dev-server-01`, click **More** and choose **Delete**.
3. Confirm deletion. Optionally release the EIP if it was dedicated.
4. Navigate to **VPC** → **Security Groups**, select `sg-xgate`, and delete it.
5. Remove the SSH config entry from `~/.ssh/config`.

> **Important**
> If you used Terraform to provision the infrastructure, run `terraform destroy` instead to remove all resources declared in the configuration.

This guide was generated using the `guide.cls` template. For the full command reference, see the template's README.md or run `/skill huawei-template-guide` in your coding agent.

7 Changelog

2.7.1 *2026-08-17*

- Rename project repository to all lowercase (`huawei-doc-templates`).

2.7.0 *2026-08-12*

- The `changelog` environment now emits its own section heading; `[nochangelog]` suppresses the heading and entries in one switch.

2.6.1 *2026-08-12*

- Fix `hutable` centering and row-color fill (switched to `\hline` for `colortbl` compatibility).

2.6.0 *2026-08-12*

- Switched verification table to the new `hutable` full-grid style.
- Floats now default to `[H]` placement (in-source order).

2.5.1 *2026-08-10*

- H1 section title size reduced from 27pt to 20pt.

2.5.0 *2026-08-10*

- Default image size increased: width 65% → 90% of text width, height 40% → 50% of text height.

2.4.0 *2026-08-10*

- Enlarged H1 section title (18pt → 27pt, 50% bigger).
- `nochangelog` option now hides version, date, and time on the cover page.

2.3.1 *2026-08-09*

- Rolled back `/ActualText` markers — broke code block rendering.

2.3.0 *2026-08-09*

- Added key-value sizing options to `\image` and `\imagecap` — `width` and `height` can now be set independently.

2.2.4 *2026-08-09*

- Fixed PDF copy-paste of URLs in code blocks — disabled Cascadia Code programming ligatures (`calt`) that corrupted `//` into `))` on copy.

2.2.3 *2026-08-09*

- Fixed italic font rendering — HarmonyOS Sans and Cascadia Code have no italic variants; enabled synthetic slant (`AutoFakeSlant`).

2.2.2 *2026-08-08*

- Fixed chapter reference in prerequisite (Chapter 3 → Chapter 4).

2.2.1 *2026-08-08*

- Changed table and figure caption labels to black (was Huawei red).

2.2.0 *2026-08-08*

- Added Huawei-branded table styling: red header bar, alternating row colors, and bordered cells.

2.1.0 *2026-08-06*

- Updated ECS flavor to `ac8.xlarge.2` (4 vCPU, 8 GB RAM).
- Updated OS image to Ubuntu 24.04 Server 64bit.
- Increased disk size to 100 GB.
- Recommended pay-per-use billing mode.
- EIP auto-assigned: pay by traffic, 1000 Mbit/s, release with ECS.
- Key pair authentication (created in Huawei Cloud); root password can be set later as a second access method.
- Security group `sg-xgate` now created **before** the ECS instance.
- Replaced SSH port 22 with port 4444 (Huawei proxy blocks port 22).
- Restricted SSH access to three specific Huawei internal IPs (`119.8.89.193`, `119.8.89.3`, `119.8.193.3`).
- Added `Port 4444` to SSH config examples.

2.0.0 *2026-08-05*

- Replaced OpenCode with oh-my-coding-maas-gateway as the default coding environment.
- Added changelog and versioning support to the template (`\setdocversion`, `\setdocdate`, `changelog` environment).
- Renamed `info` environment to `infobox` to avoid package name collisions.
- Fixed external file code block implementation (`\codefile`).

1.0.0 *2026-08-05*

- Initial version.
- Added ECS provisioning on Huawei Cloud.
- Added SSH with proxy configuration for VS Code Remote-SSH.
- Replaced OpenCode with oh-my-coding-maas-gateway (LiteLLM proxy with load balancing, virtual keys, and Grafana observability).
- Added Huawei document templates project setup.
- Added changelog and versioning support to the template.